

INTRODUCTION

Emotion Regulation

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Emotion regulation (ER) refers to attempts to influence emotions in ourselves or others. Over the past several decades, ER has become a popular topic across many subdisciplines within psychology. One framework that has helped to organize work on ER is the process model of ER, which distinguishes 5 families of strategies defined by when they impact the emotion generation process. The process model embeds these ER strategies in stages in which a need for regulation is identified, a strategy is selected and implemented, and monitoring occurs to track success. Much of the research to date has focused on a strategy called cognitive reappraisal, which involves changing how one thinks about a situation to influence one's emotional response. Reappraisal is thought to be generally effective and adaptive, but there are important qualifications. In this article, we use reappraisal as an example to illustrate how we might consider 4 interrelated issues: (a) the consequences of using ER, either when instructed or spontaneous; (b) how ER success and frequency are shaped by individual and environmental determinants; (c) the psychological and neurobiological mechanisms that make ER possible; and (d) interventions that might improve how well and how often people use ER.

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The idea that we can change our emotions (rather than just passively experience them) goes back thousands of years (Grube & Reeve, 1974). In the modern era, this idea has been expressed in foundational theoretical and empirical work on psychological defenses (Freud, 1946) and stress and coping (Lazarus & Folkman, 1984). Then in the 1990s, the field of emotion regulation (ER) began to emerge as a distinct research domain (Campos, Campos, & Barrett, 1989; Gross & Muñoz, 1995; Izard, 1990; Thompson, 1994).

ER focuses on people's attempts to influence emotions, defined as time-limited, situationally bound, and valenced (positive or negative) states. Compared with historical predecessors, ER is

broader in that it is not limited to the down-regulation of negative emotions (fear, anxiety, stress) but encompasses up- and down-regulation of positive and negative emotions in accordance with regulation-related goals. Whereas goal pursuit (and therefore ER) is often conscious and deliberate, it can also occur implicitly, outside conscious awareness (Braunstein, Gross, & Ochsner, 2017; Koole, Webb, & Sheeran, 2015). To date, regulation of one's own emotions (intrinsic regulation) has been a primary focus of research, but there is increasing interest in the regulation of someone else's emotions (extrinsic regulation; Nozaki & Mikolajczak, 2020; Zaki & Williams, 2013) as well as regulating emotions in group contexts (Goldenberg, Halperin, van Zomeren, & Gross, 2016; Porat, Tamir, & Halperin, 2020).

In this article, we first outline the process model of ER, which distinguishes five families of ER strategies and describes four stages by which ER strategies are enacted. We use one well-studied and frequently used ER strategy, cognitive reappraisal, as an example throughout, offering clarifying comparisons with other strategies. We review the consequences of using reappraisal and then consider the contextual and individual determinants (moderators) that influence how well and how often reappraisal is used. We next review psychological and neurobiological mechanisms (mediators) of reappraisal. Finally, we review interventions designed to improve individuals' emotional lives by increasing how well or how often reappraisal is used.

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What Is Emotion Regulation?

ER researchers seek to identify the different ways that individuals engage in ER and determine which ER strategies are most effective (successful at achieving an ER goal) and adaptive (associated with better long-term mental and physical health outcomes). The process model of emotion regulation (Gross, 2015) is one helpful organizational scheme.

At the first level of this model (Figure 1a), emotion generation is sequentially described as encountering relevant situations, attending to key aspects of those situations, appraising the situations in relation to active goals, and having experiential, physiological, and/or behavioral responses. For example, an individual might have a job interview (situation), notice the cool demeanor of the evaluator (attention), interpret the coolness as displeasure with the interviewee (appraisal), and experience fear, shortness of breath, and begin fidgeting (response) in the interview. The feedback arrow in Figure 1a indicates that the output of the emotion gener-

ation cycle creates a new aspect of the situation (the situation is now sitting in a job interview while feeling afraid, being short of breath, and fidgeting), and the cycle repeats.

What kinds of things do people do when they want to influence emotion? ER strategies can be organized into five families according to the emotion generation stage at which they first intervene (see Table 1 and Figure 1b). Table 1 outlines the five families, gives nonexhaustive example strategies for each family, and includes specific tactics which might be implemented in the context of the job interview scenario.

Given that people can use many different ER strategies, what processes govern their use of these strategies? According to the process model (Gross, 2015), the ER cycle begins with a discrepancy between someone's goal state (i.e., the emotional state they desire) and the actual (or projected) state. This discrepancy is then identified as an opportunity for regulation (1), a regulation strategy is selected from alternatives (2), the

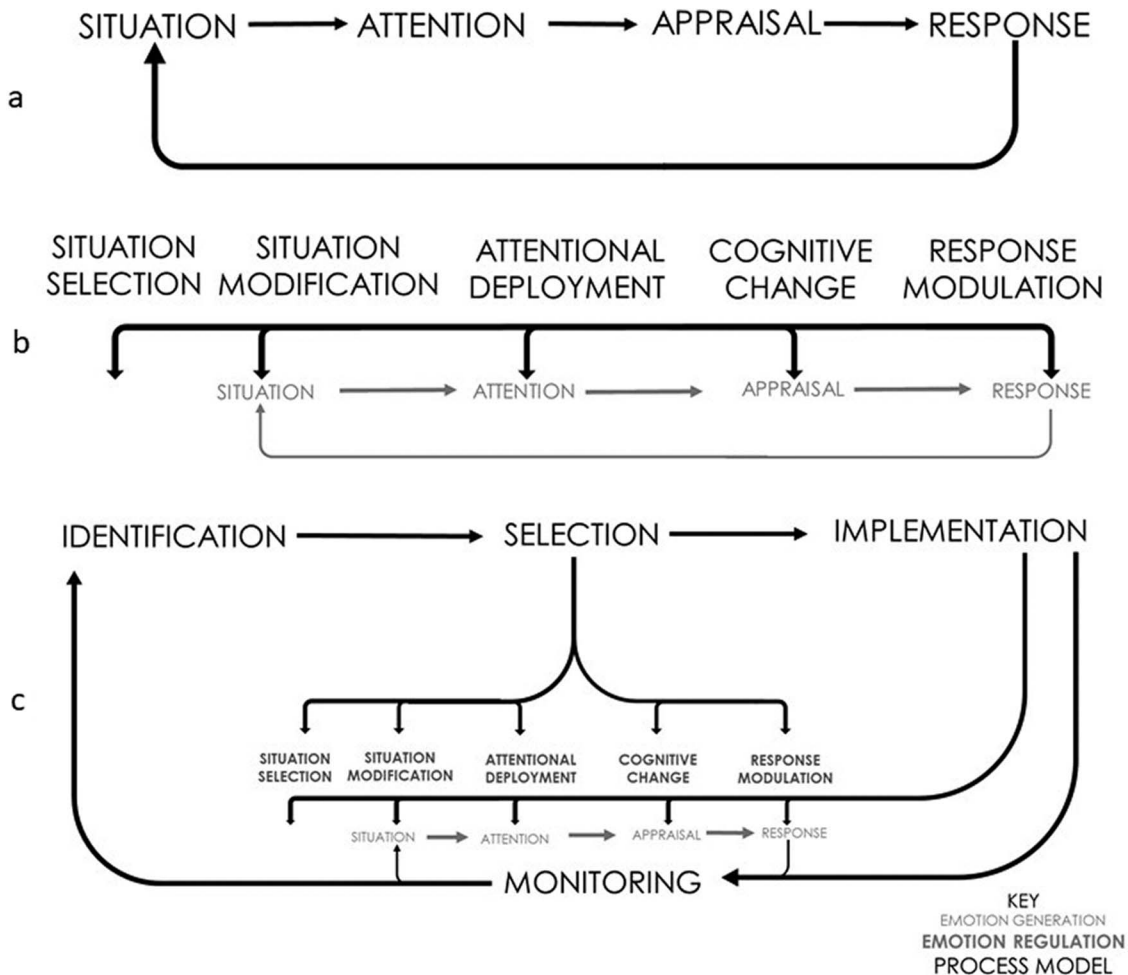


Figure 1. A sequential model of emotion generation (a), the five families of strategies that can be used to regulate emotions, organized by the stage of emotion generation in which they first intervene (b), and the process model of emotion regulation, outlining four stages by which emotion regulation strategies are enacted (c). Figures are modified from Uusberg et al. (2019). Feedback arrows indicate that all three stages are constantly iterating cycles.

Table 1

The Five Families of ER Strategies, Strategy, and Tactic Examples From the Job Interview Context, Organized by the Stage in the Emotion Generation Cycle in Which They First Intervene

Strategy family	Selected strategies	Example tactics (in job interview context)
Situation selection	Avoidance	Declining initial engagement with emotional situations (e.g., declining interview request)
Situation modification	Direct request	Taking action to influence a situation once engaged (e.g., asking to move interview to a private location)
Attentional deployment	Distraction	Directing attention (internal or external) away from the emotional situation toward nonemotional aspects of the situation or nonemotional situations (e.g., glancing at notes rather than the interviewer's face)
	Rumination	Recurrently directing attention toward causes and consequences of emotion (e.g., mentally replaying a moment in which the interviewer appeared to frown at your response)
Cognitive change	Cognitive reappraisal	Reinterpreting or reevaluating the emotional situation and/or your goals (e.g., telling yourself the interviewer was likely coached to not give overt positive feedback to applicants)
	Acceptance	Welcoming emotions with nonevaluative judgment (e.g., noticing and cultivating curiosity about manifestations of anxiety in the interview)
Response modulation	Expressive suppression	Preventing outward expression of internal emotional state (e.g., keeping face neutral to avoid expressing disappointment)
	Physiological intervention	Directly altering emotion-relevant physiology using actions or substances (e.g., slowing one's breathing rate)

Note. ER = emotion regulation.

strategy is implemented through specific tactics (3), and the whole cycle is monitored for success in achieving the regulatory goal (4; [Figure 1c](#)). In the job interview scenario, identification would involve noticing the feeling of fear but wanting to feel excited, paired with believing it is possible for this emotion to change. Selection would occur when deciding to modify expressive behavior rather than attention to the interviewer. Implementation would involve increasing tension on facial muscles to prevent displaying a worried countenance. Monitoring could include asking, “Am I feeling more excited because I decided to not display my worry?” as well as noting changes in the environment, which would indicate the need to continue, stop, or switch to a different ER strategy—processes that contribute to ER flexibility ([Aldao, Sheppes, & Gross, 2015](#); [Bonnano & Burton, 2013](#); [Pruessner, Barnow, Holt, Joormann, & Schulze, 2020](#)).

What Consequences Does Emotion Regulation Have?

Most studies of emotion regulation have either manipulated ER or assessed typical patterns of ER use ([McRae, 2013](#)). The consequences of manipulating ER (implementation stage) have typically been the purview of basic affective science because strategies are trained and cued in the laboratory, and the measured emotional outcomes are temporally proximal to the regulation. Researchers often refer to this as ER effectiveness, ability, capacity, or success (which we use here). Initial studies of ER success tested the hypothesis that the timing of the ER strategy, in which timing can refer either to relative timing within one cycle or timing of earlier versus later repeating cycles, predicted ER success ([Gross, 2001](#); [Sheppes & Gross, 2011](#)). By contrast, patterns of ER use (selection stage) have largely been studied within personality, developmental, and clinical psychology, are most typically measured using questionnaires, and many of the presumed consequences are relatively distal associates. Typical ER use is variously referred to as ER tendency, ER use, habitual ER, trait ER, or ER frequency (which we use here). In the last decade, ER frequency has increas-

ingly been operationalized as how often someone chooses to use a particular strategy in the laboratory ([Sheppes et al., 2011](#)).

Reappraisal is frequently successful, inasmuch as it often results in the desired changes in self-reported emotion ([Gross, 1998](#); [Jackson, Malmstadt, Larson, & Davidson, 2000](#)), peripheral physiology ([Denson, Grisham, & Moulds, 2011](#); [Dillon & LaBar, 2005](#); [Ray, McRae, Ochsner, & Gross, 2010](#)), and neural measures of emotion ([Chang, Gianaros, Manuck, Krishnan, & Wager, 2015](#); [Dörfel et al., 2014](#); [Hajcak & Nieuwenhuis, 2006](#); [Kalisch et al., 2005](#); [Lohani & Isaacowitz, 2014](#); [Ochsner, Bunge, Gross, & Gabrieli, 2002](#); [Schaefer et al., 2002](#); [Shahane, Lopez, & Denny, 2019](#)). Studies using experience sampling to examine reappraisal in more natural environments have been largely consistent with laboratory findings ([Colombo et al., 2020](#)). Reappraisal's success contrasts with suppression, which results in weak, null, or paradoxical (reversed) changes in negative emotion ([Goldin, McRae, Ramel, & Gross, 2008](#); [Gross, 1998](#); [Kalokerinos, Greenaway, & Denson, 2015](#)). Consequences of reappraisal also contrast which distraction, in that reappraisal results in less short-term success in decreasing negative emotion than distraction, but is more effective upon later encounters with the stimulus ([Hermann, Kress, & Stark, 2017](#); [Thiruchselvam, Blechert, Sheppes, Rydstrom, & Gross, 2011](#); for a meta-analysis, see [Webb, Miles, & Sheeran, 2012](#)). Factors that lessen reappraisal's success are discussed below in the determinants section (for a review, see [Ford & Troy, 2019](#)).

Greater reappraisal frequency is often associated with adaptive outcomes such as greater physical health ([Appleton, Buka, Loucks, Gilman, & Kubzansky, 2013](#); [Appleton, Loucks, Buka, & Kubzansky, 2014](#)), higher academic achievement ([Davis & Levine, 2013](#); [Ivcevic & Brackett, 2014](#)), more positive social outcomes ([English, John, Srivastava, & Gross, 2012](#)), greater psychological well-being, ([Gross & John, 2003](#)), and fewer symptoms psychopathology ([Aldao, Nolen-Hoeksema, & Schweizer, 2010](#); [Cludius, Mennin, & Ehring, 2020](#)). These associations contrast with greater suppression frequency, which is often associated with lesser well-being, more symptoms of psychopathology, and lesser relationship satisfaction ([Cameron & Overall, 2018](#); [Chervonsky & Hunt,](#)

2017; English et al., 2012; English & Eldesouky, 2020; Gross & John, 2003). However, suppression seems to be a relatively adaptive skill early in development, and more suppression use is associated with greater school readiness in preschoolers (Harrington, Trevino, Lopez, & Giuliani, 2020).

What Determines (Moderates) Emotion Regulation?

Because ER strategies have different consequences, what determines how well or how often people use different ER strategies? To address these questions, researchers have identified individual and environmental factors that moderate the ER success or frequency.

Reappraisal's success varies by context: Laboratory studies indicate that reappraisal is more successful when the negative emotion is of moderate (not high) intensity (Shafir, Schwartz, Blechert, & Sheppes, 2015; but see Silvers, Weber, Wager, & Ochsner, 2015), is generated from cognitive rather than perceptual emotional stimuli (McRae, Misra, Prasad, Pereira, & Gross, 2011; Nelson, Fitzgerald, Klumpp, Shankman, & Phan, 2015), and when there is relatively more time available to regulate (Kalokerinos, Résibois, Verduyn, & Kuppens, 2017; Sheppes & Meiran, 2007). The presence of positive emotion, perhaps only when related to the negative emotion-eliciting situation, may facilitate reappraisal success (Waugh, 2020). The engagement of prefrontal cortex (PFC) control regions during reappraisal (Buhle et al., 2013; Morawetz, Bode, Derntl, & Heekeren, 2017; Ochsner & Gross, 2008) suggests that reappraisal may be less successful under conditions that impair PFC-dependent cognition. Indeed, sleep deprivation (Walker & van der Helm, 2009), poor-quality sleep (Mauss, Troy, & LeBourgeois, 2013), stress (Raio, Oederu, Palazzolo, Shurick, & Phelps, 2013; Zhan et al., 2017), and developmental variation in PFC integrity and functioning (DeCicco, Solomon, & Dennis, 2012; McRae et al., 2012) are associated with less successful reappraisal.

What factors determine how often people use reappraisal? Reappraisal frequency is less heritable and more open to nonfamilial environmental influences than emotion-relevant personality dimensions or suppression frequency (McRae et al., 2017). Contextual and individual factors that determine ER frequency include social relationship partners, especially in childhood (Lindsey, 2020), and personality factors (Hughes, Kratsiotis, Niven, & Holman, 2020). Individuals select reappraisal more frequently when the stimuli to be regulated are of lesser intensity (Doré, Weber, & Ochsner, 2017; Martins, Sheppes, Gross, & Mather, 2016; Opitz, Cavanagh, & Urry, 2015) or contain more reappraisal affordances (Suri et al., 2017) and use reappraisal less frequently when they have anticipatory information about the content (but not timing) of emotion-relevant events (Shafir & Sheppes, 2020). When stimuli are of high intensity, distraction (Sheppes, Scheibe, Suri, & Gross, 2011) or suppression (Dixon-Gordon, Aldao, & De Los Reyes, 2015) are more likely to be used. Individuals from cultures that value self-reflection and insight tend to use reappraisal more frequently than average (Haga, Kraft, & Corby, 2009), whereas individuals from cultures that value open expression of emotion tend to use suppression less frequently (Matsumoto, Yoo, & Nakagawa, 2008; McRae, Heller, John, & Gross, 2011; Su et al., 2015). Furthermore, there is evidence that a history of having short-term success might lead to more frequent use of maladaptive

strategies such as self-injurious behavior (Swerdlow, Pearlstein, Sandel, Mauss, & Johnson, 2020).

What Are the Mechanisms (Mediators) of Emotion Regulation?

What neurobiological and psychological mechanisms enable reappraisal? Neurobiological mechanisms of reappraisal implementation include brain systems that support cognitive control and linguistic elaboration, compared with distraction, which uses more external attentional control systems, and suppression, which uses more inhibitory systems (Goldin et al., 2008; Hayes et al., 2010; Kanske, Heissler, Schönfelder, Bongers, & Wessa, 2011; McRae et al., 2010; Ohira et al., 2006; Vanderhasselt, Kühn, & De Raedt, 2012). Engagement of reappraisal-related prefrontal and parietal cognitive control and linguistic elaboration systems can lead to either diminished or enhanced emotional responding, in correspondence with one's emotional goal (Ochsner et al., 2004; Kim & Hamann, 2007). Clarifying how reappraisal relates to affective versus cognitive control may clarify the developmental trajectory of ER during adolescence (Schweizer, Gotlib, & Blakemore, 2020). Likewise, clarifying whether reappraisal can be driven by proactive versus reactive control processes may help us determine the most effective and adaptive ER strategies to use in adulthood and later in life (Martins-Klein, Alves, & Chiew, 2020). Another potential mechanism is the (up-)regulation of positive emotion, which may help us better understand associations between ER and psychopathology (Silton et al., 2020).

Psychological mechanisms governing the selection of reappraisal include decision-making processes, in which the need for regulation is balanced with anticipated success, the estimated cognitive costs of implementing candidate strategies (Milyavsky et al., 2019), and the desire to engage with the emotional aspects of the situation to be regulated (Sheppes et al., 2014). The process of reappraisal selection may engage a similar frontoparietal network to that engaged during reappraisal implementation, but few studies are able to separate mechanisms of selection from implementation (Cosme, Mobasser, Zeithamova, Berkman, & Pfeifer, 2018).

Which Interventions Improve Emotion Regulation?

Psychosocial interventions that target ER to treat psychopathology are common (Berking et al., 2008; Rottenberg & Gross, 2007) and have considerable empirical support (Butler, Chapman, Forman, & Beck, 2006). Interventions can target neurobiological or psychological mechanisms of ER and measure their proximal or distal effects. Interventions for children often cast a wide net by educating parents and/or teachers about healthy ER and offering concrete steps for children to observe and practice ER either through parent socialization at home (England-Mason & Gonzalez, 2020) or instruction about emotional intelligence (which includes emotion regulation) at school (Hoffman, Brackett, Bailey, & Willner, 2020).

Less frequent use of largely adaptive strategies like reappraisal and, to a greater extent, overuse of maladaptive strategies like rumination and suppression consistently characterize clinical groups with mood disorders (Aldao et al., 2010; Compas et al., 2017; Potthoff et al., 2016). The goal of many interventions is to decrease the frequency with which maladaptive ER strategies are used and to

increase the frequency and success with which adaptive ER strategies are used (Aldao, Jazaieri, Goldin, & Gross, 2014; Brozovich et al., 2015; Gilboa-Schechtman, Shachar, & Sahar, 2014). Cognitive therapies, including cognitive-behavioral therapy, directly target reappraisal skills (Beck & Dozois, 2011; Kazantzis et al., 2018). Cognitive-behavioral therapy improves self-reported reappraisal success (Goldin et al., 2014), and other interventions that improve reappraisal success include direct reappraisal training using a picture-based task (Denny, 2020). Neurobiological interventions such as antidepressant medication increase reappraisal frequency (McRae, Rekshan, Williams, Cooper, & Gross, 2014) and success (Outhred et al., 2015), and noninvasive stimulation of neural regions engaged during reappraisal increases reappraisal success (Feesser, Prehn, Kazzer, Mungee, & Bajbouj, 2014) and decreases symptoms of depression (Lan, Chhetry, Liston, Mann, & Dubin, 2016).

Directions for Future Research

To round out this discussion, we will first briefly review research that falls outside the selection and implementation stages of the process model. We will then turn to a high-stakes example that demonstrates the importance of separating implementation from selection.

So far, we have focused on implementation success and selection frequency in the context of reappraisal. What about the consequences, determinants, mechanisms, and interventions that influence the identification and monitoring stages? The consequence of not identifying a need to regulate is emotion regulation failure, and identification appears to be jointly determined by an individual's emotion goals (which can be hedonic or instrumental; Tamir, Mitchell, & Gross, 2008) and their belief in the malleability of emotion (Tamir, John, Srivastava, & Gross, 2007). Mechanisms of identification include processes that govern goal pursuit, including goal setting, goal striving (Tamir, Vishkin, & Gutentag, 2020), or the use of implementation intentions (Gallo & Gollwitzer, 2007). Interventions for identification target the valuation of specific emotional states (Crum, Jamieson, & Akinola, 2020) or beliefs about emotion malleability (Kneeland, Nolen-Hoeksema, Dovidio, & Gruber, 2016). Although monitoring is an important stage of the process model, few studies have examined consequences, determinants, mechanisms, and interventions related to monitoring of ER specifically (but see Koch, Mars, Toni, & Roelofs, 2018). One exciting exception is the successful externalization of monitoring in functional magnetic resonance imaging-based neurofeedback (Paret & Hendlar, 2020). In future research on emotion regulation, it will be important to examine the full range of stages in the emotion regulation process.

Throughout this discussion, we have distinguished between how well and how often people use reappraisal. Why is this important? In many circumstances, they are conflated, especially as researchers, including ourselves, briefly summarize findings of previous studies of ER. This conflation is reasonable because the short-term success of reappraisal (e.g., successfully reduced negative emotion) could logically lead to some of the long-term adaptive associates of reappraisal frequency. (e.g., lower levels of daily negative affect). However, there is reason to believe how well and how often reappraisal is used might be distinct constructs (McRae, 2013; Silvers & Moreira, 2019), which is especially apparent in clinical contexts.

One of the most reliable findings in the ER literature is that greater use of reappraisal is related to fewer symptoms of psychopathology (Aldao et al., 2010; Eftekhari, Zoellner, & Vigil, 2009; Hu et al., 2014), but studies of how well reappraisal is used by clinical groups have reported weak or null differences from non-clinical groups. With few exceptions, it appears that members of clinical groups can successfully use reappraisal in a laboratory setting (Goldin, Manber, Hakimi, Canli, & Gross, 2009). Meta-analyses indicate that members of clinical groups demonstrate reappraisal success that is indistinguishable from controls in 80% of published studies (Picó-Pérez, Radua, Steward, Menchón, & Soriano-Mas, 2017; Zilverstand, Parvaz, & Goldstein, 2017). What is the source of this disconnect between reappraisal success and use?

One possibility is that although individuals from clinical groups can use reappraisal successfully when cued, they fail to appropriately identify moments at which ER would be helpful in everyday life. Alternatively, members of clinical groups may in fact be able to identify moments at which ER would be helpful but for one reason or another choose not to use reappraisal very frequently in everyday life (as implied by Dryman & Heimberg, 2018). It is also possible that laboratory use measures capacity, which is an overestimate of actual success everyday life. Finally, it is possible that this disconnect between reappraisal use and success is an artifact of the way these constructs are measured (cumulative emotion ratings on a laboratory task vs. self-report responses). The source of the disconnect could have important implications for ER intervention science, which would respectively focus on using reminders and encouragement to select and initiate reappraisal in everyday life, improving conditions for implementing reappraisal in everyday life, or developing, refining, or combining measures of reappraisal that most closely correspond to documented emotional difficulties in clinical groups.

These suggestions for future research are, of course, a small sampling of the full set of possibilities currently evident in this fast-growing field. Our goal is to show that research on ER has produced sharpened definitions and generative models of ER, outlined different emotional consequences of engaging in different types of ER, identified moderating contextual and individual factors that impact ER, described psychological and neurobiological mechanisms by which regulation influences emotion, and documented the effect of ER interventions on short- and long-term outcomes. Additional reading beyond the scope of our discussion can be found in [online supplemental materials](#). In our view, when considering (or conducting) research on ER, it is helpful to clarify which stage of the ER process (identification, selection, implementation, or monitoring) is being described, manipulated, or measured because this will help to make subsequent ER interventions more targeted and precise. Whereas we have focused primarily on cognitive reappraisal here, research on other ER strategies is obviously critical to clarify the most adaptive and effective ways to influence emotions, with the ultimate goal of finding better ways to achieve better emotional lives.

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